

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE CODE: MLA 0202

COURSE NAME: Fundamentals Of Machine Learning

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COURSE OUTCOME COVERED

CO3: Analyse unsupervised learning methods, including clustering and dimensionality reduction.

CO4: Develop probabilistic graphical models for inference, learning patterns, and analysing sequential data.

CO5: Create graphical models to analyse and infer patterns in complex probabilistic systems for structured decision-making.

Total Marks:100

Code of Conduct:

I G C Chandravardhan Reddy-192425037 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G Chandravardhan Reddy

Particular	Details
Department:	AIML
Programme:	FACULTY OF ENGINEERING
Course Code & Course Name	MLA02 Fundamentals of Machine Learning
Academic Year / Batch	2026-2027
Faculty Name	Dr.M.S.Saravanan
Assignment Title	Design and Development of a Probabilistic Graphical-Model-Based Patient Cohort Segmentation and Clinical-State Reasoning System
Date of Issue	01-09-2026
Date of Submission	04-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO3: Analyse unsupervised learning methods, including clustering and dimensionality reduction. CO4: Develop probabilistic graphical models for inference, learning patterns, and analysing sequential data. CO5: Create graphical models to analyse and infer patterns in complex probabilistic systems for structured decision-making.
Bloom's Taxonomy Level	L4-Analyse, L5-Evaluate, L6-Create
SDG Mapping (SDG 1-17)	SDG 3 – Good Health and Well-being SDG 9 – Industry, Innovation and Infrastructure SDG 10 – Reduced Inequalities
Industry / Societal Relevance	Enables AI personalized healthcare by segmenting patient risk groups & predicting disease progression to optimize clinical decision-making & improve patient outcomes.

B. Assignment Problem / Challenge

Design and Development of a Probabilistic Graphical-Model-Based Patient Cohort Segmentation and Clinical State Reasoning System.

C. Problem Statement

A multi-speciality hospital wants to deploy an analytics module that **segments patients into clinically meaningful risk cohorts** from their vitals and laboratory profile, and **reasons about the probable cause of an abnormal reading and a patient's evolving clinical state** using probabilistic graphical models. The existing hospital workflow relies on manual review of high-dimensional patient data, making cohort identification, dependency reasoning, and clinical-state tracking **slow, inconsistent, and difficult to explain**.

Students are required to analyze the given case and design an **unsupervised patient-cohorting and probabilistic-graphical-model-based reasoning pipeline**. The solution should include a clustering-based cohort-discovery step, a dimensionality-reduction step for the high-dimensional vitals/lab feature space, a Bayesian network for reasoning about symptom–risk–disease dependencies, and a Hidden Markov Model for tracking a patient's evolving clinical state.

The pipeline should support operations such as **segmenting patients using K-means and/or Gaussian Mixture/EM clustering, reducing dimensionality using PCA or Factor Analysis, representing symptom–risk–outcome dependencies using a Bayesian network** with exact inference to compute posterior probabilities given observed evidence, and **modelling a patient's sequence of clinical states** (e.g., Stable → Deteriorating → Critical) as a Hidden Markov Model, using the forward algorithm or Viterbi decoding to estimate the most likely current state. The graphical-model outputs should feed into a **structured decision rule** (e.g., escalate/monitor/discharge) for clinical or resource-related decision making.

The design should consider requirements such as **cohort interpretability, justified retained variance/components, correct probabilistic inference, explainability, and data privacy**. Students should justify their design decisions and demonstrate the working pipeline using an

appropriate toolset such as **Python, scikit-learn, pgmpy, or hmmlearn.**

D. Requirements and Constraints

1. Functional Requirements

The functional requirements define the major functions that the proposed patient-risk analysis system must perform.

1.1 Patient Risk Cohort Segmentation

- The system shall segment patients into a small number of meaningful and interpretable risk cohorts using clustering algorithms such as **K-Means** and **Gaussian Mixture Models (GMM)**.
- The clustering process shall use relevant patient features such as vital signs, laboratory measurements, and clinical observations.
- The identified groups may be interpreted as **Low Risk, Moderate Risk, and High Risk** based on their clinical characteristics.
- K-Means shall provide hard cluster assignments, while GMM shall provide probabilistic or soft cluster assignments.

1.2 Dimensionality Reduction

- The system shall reduce high-dimensional patient data using **Principal Component Analysis (PCA)** or **Factor Analysis**.
- The dimensionality-reduction process shall preserve the important structure and information contained in the original features.
- PCA shall be used to transform multiple correlated clinical variables into a smaller number of principal components.
- The reduced feature space shall support efficient analysis, clustering, and visualization.

1.3 Bayesian Network-Based Reasoning

- The system shall represent relationships between symptoms, risk factors, and disease

outcomes using a **Bayesian Network**.

- The dependencies shall be represented using a **Directed Acyclic Graph (DAG)**.
- The system shall calculate posterior probabilities based on observed patient evidence.
- The system shall support probability queries such as:

$P(\text{Disease} \mid \text{Observed Evidence})$

- Bayesian inference shall help estimate the probability of a disease or severe-risk condition based on available clinical evidence.

1.4 Clinical State Tracking

- The system shall track the patient's underlying clinical condition over time using a **Hidden Markov Model (HMM)**.
- The model shall represent possible clinical states such as:

Stable -> Deteriorating -> Critical

- Patient vital signs and other sequential observations shall be used as evidence for estimating the hidden clinical state.
- The **Forward algorithm** may be used to calculate state probabilities, while the **Viterbi algorithm** may be used to determine the most probable sequence of clinical states.

1.5 Decision Recommendation

- The system shall translate the outputs of clustering, Bayesian inference, and HMM state prediction into a structured recommendation.
- The primary recommendations shall include:

Escalate

Monitor

Discharge

- The decision logic shall combine the estimated severe-risk probability and the predicted clinical state.

- The recommendation shall be presented as **decision support** and shall not replace professional clinical judgment.

2. Performance Requirements

The performance requirements specify the expected efficiency, responsiveness, and predictive reliability of the system.

2.1 Computational Efficiency

- The clustering algorithms shall process the available patient dataset efficiently.
- PCA or Factor Analysis shall reduce the computational complexity associated with high-dimensional clinical data.
- The system should avoid unnecessary processing and memory usage during model training and prediction.
- The complete analysis pipeline should be suitable for the available academic computing environment.

2.2 Response Time

- Bayesian Network inference should produce posterior probabilities within a few seconds for an individual patient query.
- HMM state inference should also be completed within a few seconds for a normal patient observation sequence.
- The combined prediction and decision process should provide results within a reasonable response time for interactive use.

2.3 High-Risk Case Detection

- The system shall aim to maintain a low false-negative rate for high-risk and critical cases.
- A **false negative** occurs when a genuinely high-risk patient is incorrectly classified as low or moderate risk.
- Since missing a critical case may have serious consequences, the system should prioritize **sensitivity or recall** for high-risk cases.

- A higher false-positive rate may be acceptable if it helps reduce the number of missed critical cases.

3. Technical Constraints

Technical constraints define the technologies, data structures, algorithms, and implementation environment that restrict the system design.

3.1 Patient Data Representation

- Patient records shall be represented as numerical feature vectors.
- Features may include vital signs, laboratory measurements, and other relevant clinical variables.
- Categorical data shall be appropriately encoded before being provided to clustering or dimensionality-reduction algorithms.
- Numerical features shall be appropriately scaled or standardized before model training.

Example:

Patient = [Age, Heart Rate, Blood Pressure, Glucose, Temperature, SpO2]

3.2 Bayesian Network Constraints

- Clinical dependencies shall be represented using a **Directed Acyclic Graph (DAG)**.
- The Bayesian Network shall use **Conditional Probability Tables (CPTs)** to represent conditional relationships between variables.
- The graph shall not contain directed cycles.
- The network shall support probabilistic inference using appropriate inference methods such as **Variable Elimination**.

3.3 Hidden Markov Model Constraints

- Sequential patient observations shall be modeled using a **discrete-state Hidden Markov Model**.
- The model shall contain:

- Initial state distribution
- State transition probabilities
- Observation/emission probabilities
- The primary hidden states shall represent different levels of clinical condition, such as:

Stable

Deteriorating

Critical

- The system shall use the Forward algorithm for probability estimation and/or the Viterbi algorithm for most-probable state-sequence decoding.

3.4 Software and Technology Constraints

The implementation shall use open-source technologies:

Python

scikit-learn

pgm-py

hmmlearn

Their primary purposes are:

Technology	Purpose
Python	Main development language
scikit-learn	Preprocessing, K-Means, GMM, PCA
pgmpy	Bayesian Network construction and inference

hmmlearn	Hidden Markov Model implementation
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4. Time and Resource Constraints

4.1 Academic Timeline

- The complete system shall be designed and implemented within the assigned academic project timeline.
- The development process shall include data preparation, model development, integration, testing, evaluation, and documentation.
- The implementation shall prioritize the core requirements before optional features.

4.2 Hardware Requirements

- The system shall be capable of running on standard laboratory or personal-computer hardware.
- Specialized computing resources such as high-end GPUs or dedicated servers shall not be required for the basic implementation.
- The algorithms should be selected and configured to operate efficiently on the available hardware.

4.3 Dataset Constraints

- The system shall use **synthetically generated or properly de-identified datasets**.
- Directly identifiable patient information shall not be required for development or testing.
- The dataset should contain sufficient clinical features to demonstrate clustering, Bayesian reasoning, and sequential state tracking.

5. Reliability, Safety, and User Requirements

5.1 Reliability

- The system shall provide consistent results when the same input and model configuration are used.
- Data preprocessing and model-prediction procedures shall follow the same defined pipeline for training and testing.
- The system should be evaluated using suitable performance measures such as accuracy, precision, recall, F1-score, and false-negative rate.

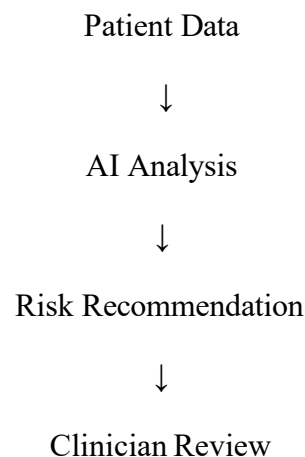
5.2 Safety

- The system shall prioritize the detection of high-risk and critical conditions.
- False negatives should be minimized even if this results in a higher number of false positives.
- Risk thresholds shall be treated as project-defined decision-support rules rather than independently validated clinical guidelines.

5.3 Clinician-in-the-Loop

- The system shall operate strictly as a **decision-support system**.
- Automated predictions shall not independently determine patient treatment or final clinical outcomes.
- A qualified healthcare professional shall review the system's recommendation before making a final decision.

The workflow shall be:





Final Decision

5.4 Explainability

- The system shall provide understandable information supporting its recommendations.
- Users should be able to identify the major factors contributing to the predicted risk.
- The system should display relevant outputs such as:
 - Risk probability
 - Patient risk cluster
 - Predicted HMM state
 - Final recommendation

Example:

Risk Probability = 0.78

Clinical State = Critical

Risk Cohort = High Risk

Recommendation = Escalate

This makes the output easier for clinical staff and administrators to understand and evaluate.

6. Security, Privacy, and Ethical Considerations

6.1 Data Privacy

- Sensitive patient information shall be protected throughout data collection, storage, processing, and analysis.
- Patient datasets should be **de-identified** before being used for system development.
- Personally identifiable information should not be included unless specifically required and appropriately protected.

6.2 Data Security

- Patient data shall be stored using appropriate access controls.
- Only authorized users should have access to sensitive information.
- Data should not be unnecessarily shared, copied, or exposed during system development.

6.3 Algorithmic Fairness

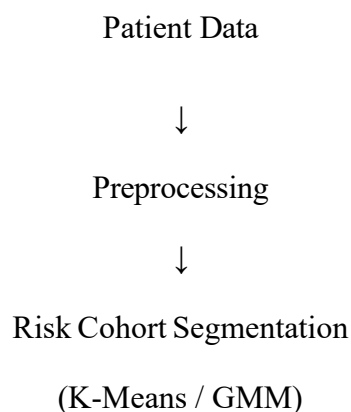
- The system shall be evaluated for potential algorithmic bias.
- Model performance should be assessed across relevant demographic groups where suitable data is available.
- Differences in false-positive and false-negative rates should be examined to identify potential unfairness.
- The objective is to ensure that the system provides consistent and fair decision-support performance across different patient groups.

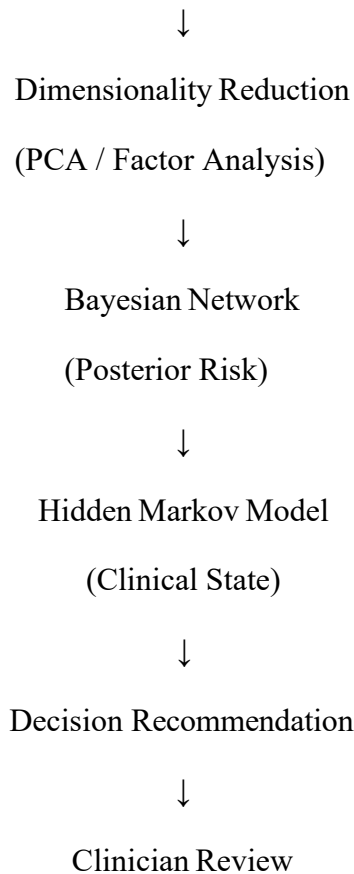
6.4 Ethical Use

- The system shall be used only as an **assistive decision-support tool**.
- Automated recommendations shall not be treated as definitive medical diagnoses.
- Human oversight shall be maintained throughout the decision-making process.
- The limitations and uncertainty of the models should be clearly communicated to users.

Summary of Requirements

The proposed system must satisfy the following overall requirements:





Overall, the **Requirements and Constraints** define a system that is capable of identifying patient risk groups, reducing complex clinical data, performing probabilistic reasoning, tracking health conditions over time, and generating explainable decision-support recommendations while operating within defined **performance, technical, resource, safety, privacy, and ethical constraints**.

E. Proposed Work

1. Problem Understanding and Formulation

Problem Identification: Manual analysis of high-dimensional vitals and laboratory profile data in clinical settings is slow, error-prone, inconsistent, and lacks algorithmic explainability. This makes early cohort identification and tracking evolving patient states difficult for healthcare staff.

Expected Outcomes: An integrated machine-learning pipeline that automatically:

- Clusters patients into interpretable risk cohorts.
- Reduces redundant features using dimensionality reduction.

- Computes disease/risk posterior probabilities using a Bayesian Network.
- Decodes evolving patient clinical states over time using a Hidden Markov Model (HMM).
- Translates outputs into a structured recommendation rule (e.g., *Escalate / Monitor / Discharge*).

Available Data: Synthetic/de-identified patient records containing vitals (heart rate, blood pressure, oxygen saturation), laboratory results (WBC, serum creatinine, lactate), demographic attributes, and time-series vital sequences.

Assumptions:

- Vital-sign measurements contain zero-mean Gaussian measurement noise.
- Clinical probability distributions in Conditional Probability Tables (CPTs) accurately reflect clinical risk dynamics.
- Patient risk cohorts remain reasonably stable over fixed evaluation windows.

Constraints: Prioritizes high model interpretability, exact probabilistic inference, explainable rules, strict patient data privacy, and execution using open-source Python toolkits within specified compute constraints.

2. Application of Course Knowledge

Clustering Methods

1. K-Means Clustering

K-Means is used to divide patient feature vectors $X = \{x_1, x_2, \dots, x_n\}$

into K risk clusters. The objective is to minimize the within-cluster sum of squared distances (inertia):

$$J = \sum_{k=1}^K \sum_{x_i \in S_k} \|x_i - \mu_k\|^2$$

where:

- S_k = set of patients belonging to cluster k
- μ_k = centroid of cluster k

Calculation: For each patient, the Euclidean distance from every centroid is calculated. The patient is assigned to the nearest centroid. The centroids are then recalculated until

the cluster assignments become stable.

For example, if a patient's distances from three centroids are 2.1, 4.5, and 3.2, the patient is assigned to Cluster 1 because 2.1 is the minimum distance.

2. Gaussian Mixture Model (GMM)

GMM provides soft cluster assignments, meaning that a patient can have a probability of belonging to every cluster.

The probability density is:

$$P(x) = \sum_{k=1}^K \pi_k * N(x | \mu_k, \Sigma_k)$$

where:

- π_k = mixing weight of cluster k
- μ_k = mean vector
- Σ_k = covariance matrix
- N = Gaussian distribution

Using Expectation-Maximization (EM):

- **E-step:** Calculate the probability that each patient belongs to each cluster.
- **M-step:** Update the mean, covariance, and mixing weights.
- Repeat these steps until the model converges.

For example, a patient may have:

$$P(C1 | x) = 0.70$$

$$P(C2 | x) = 0.20$$

$$P(C3 | x) = 0.10$$

Therefore, the patient is most likely to belong to Cluster 1, with a probability of 70%.

Dimensionality Reduction

Principal Component Analysis (PCA)

PCA reduces the number of patient features while preserving the most important information.

After centering the feature matrix X , the covariance matrix is calculated as:

$$\Sigma = (1 / (n - 1)) X^T X$$

The eigenvalue equation is:

$$\text{Sigma} * v_k = \text{lambda_k} * v_k$$

where:

- v_k = principal component
- lambda_k = eigenvalue representing the variance explained by the component

The explained variance ratio is calculated as:

$$\text{EVR_k} = \text{lambda_k} / \text{sum}(\text{lambda_j})$$

Components are selected until the cumulative explained variance reaches at least 85%:

$$\text{sum}(\text{lambda_k}, k = 1 \text{ to } m) / \text{sum}(\text{lambda_j}) \geq 0.85$$

Example: If the first three components explain 50%, 25%, and 15% of the variance:

$$50\% + 25\% + 15\% = 90\%$$

Since 90% is greater than 85%, the first three principal components are retained.

Bayesian Networks

A Bayesian Network represents relationships between medical variables using a Directed Acyclic Graph (DAG).

Example:

Symptom -> Risk Factor -> Disease

The joint probability distribution can be represented as:

$$P(X_1, \dots, X_n) = \text{product}(i = 1 \text{ to } n) P(X_i | \text{Parents}(X_i))$$

To determine the probability of a disease based on observed evidence, Variable Elimination is used to calculate:

$$P(\text{Disease} | \text{Evidence})$$

Using Bayes' theorem:

$$P(D | E) = [P(E | D) * P(D)] / P(E)$$

For example, if:

$$P(D) = 0.10$$

$$P(E | D) = 0.80$$

$$P(E) = 0.20$$

then:

$$P(D | E) = (0.80 * 0.10) / 0.20$$

$$P(D | E) = 0.40$$

Therefore, the posterior probability of disease is **40%**.

Hidden Markov Model (HMM)

An HMM is used to model changes in a patient's health condition over time.

The hidden states are:

$S = \{\text{Stable, Deteriorating, Critical}\}$

An HMM consists of:

- **Transition matrix (A):** Probability of moving from one health state to another.
- **Emission matrix (B):** Probability of observing a symptom given a health state.
- **Initial state distribution (pi):** Probability of the initial health state.

The Viterbi algorithm determines the most likely sequence of hidden health states.

The recurrence equation is:

$$V_t(k) = \max_j [V_{t-1}(j) * A_{jk}] * B_k(y_t)$$

where:

- $V_t(k)$ = maximum probability of reaching state k at time t
- A_{jk} = transition probability from state j to state k
- $B_k(y_t)$ = probability of observing y_t in state k

Calculation: At each time step, the algorithm calculates the probability of reaching each possible state from every previous state. It selects the highest probability and stores the corresponding path. After the final observation, backtracking provides the most likely sequence of health states.

For example:

Stable -> Deteriorating -> Critical

This sequence indicates that the patient's health condition is progressively worsening.

Overall Application

The techniques work together as follows:

Patient Data -> PCA -> K-Means/GMM -> Bayesian Network -> HMM -> Risk Prediction

Thus, **PCA** reduces the dimensionality of patient data, **K-Means/GMM** identifies patient risk groups, **Bayesian Networks** estimate disease probabilities based on observed evidence, and **HMM with the Viterbi algorithm** tracks the patient's health-state progression over time.

3. Solution / Design / Methodology

Pipeline Architecture:

The proposed system follows a sequential pipeline that processes patient information from raw clinical data to final risk-based recommendations. The architecture combines data preprocessing, clustering, dimensionality reduction, Bayesian reasoning, and Hidden Markov Models for comprehensive patient-risk analysis.

1. Data Preprocessing

The first stage prepares the raw patient dataset for further analysis. Clinical data may contain laboratory measurements, vital signs, and other numerical parameters with different units and ranges.

Examples of features include:

- Blood pressure
- Heart rate
- Blood glucose
- Body temperature
- Oxygen saturation
- Cholesterol level
- Hemoglobin level

To make these features comparable, continuous variables are standardized using **Z-score normalization**.

The Z-score is calculated as:

$$z = (x - \text{mean}) / \text{standard_deviation}$$

where:

- x = observed feature value
- mean = average value of the feature
- standard_deviation = standard deviation of the feature

For example, if a patient's blood glucose is 140, the mean is 100, and the standard deviation is 20:

$$z = (140 - 100) / 20$$

$$z = 2$$

Therefore, the patient's glucose value is **2 standard deviations above the mean**.

Standardization prevents features with larger numerical values from dominating the clustering and probabilistic models.

2. Cohort Discovery

After preprocessing, the system identifies groups of patients with similar clinical characteristics. This stage uses **K-Means Clustering** or **Gaussian Mixture Models (GMM)**.

K-Means

K-Means partitions patients into K clusters based on their feature similarity.

For example:

Cluster 1 = Low Risk

Cluster 2 = Moderate Risk

Cluster 3 = High Risk

The distance between a patient and a cluster centroid can be calculated using Euclidean distance:

$$\text{distance} = \sqrt{(x_1 - c_1)^2 + (x_2 - c_2)^2 + \dots + (x_n - c_n)^2}$$

Each patient is assigned to the cluster with the smallest distance.

The centroid is then recalculated, and the process is repeated until the clusters become stable.

Gaussian Mixture Model

GMM provides a probabilistic alternative to K-Means. Instead of assigning a patient completely to one cluster, it calculates the probability of belonging to each cluster.

For example:

$$P(\text{Low Risk} \mid \text{patient}) = 0.10$$

$$P(\text{Moderate Risk} \mid \text{patient}) = 0.25$$

$$P(\text{High Risk} \mid \text{patient}) = 0.65$$

The patient is therefore most strongly associated with the **High Risk** group.

GMM is particularly useful when clinical boundaries between risk groups are not clearly separated.

3. Feature Compression

Clinical datasets may contain a large number of laboratory and vital-sign features. Using all features directly can increase computational complexity and make visualization difficult.

Therefore, **Principal Component Analysis (PCA)** is applied to reduce the dimensionality of the dataset.

The covariance matrix is calculated after centering the data:

$$\text{Sigma} = (1 / (n - 1)) X^T X$$

The principal components are obtained by solving:

$$\text{Sigma} * v_k = \lambda_k * v_k$$

where:

- Σ = covariance matrix
- v_k = principal component
- λ_k = eigenvalue associated with the component

The explained variance ratio is:

$$\text{EVR}_k = \lambda_k / \sum(\lambda_j)$$

The system selects enough principal components to retain at least 85% of the total variance.

For example:

$$\text{PC1} = 50\%$$

$$\text{PC2} = 25\%$$

$$\text{PC3} = 15\%$$

Therefore:

$$\begin{aligned} \text{Total explained variance} &= 50\% + 25\% + 15\% \\ &= 90\% \end{aligned}$$

Since 90% is greater than the required 85%, the first three components are retained.

These components can then be plotted in 2D or 3D to visualize patient clusters.

4. Dependency Reasoning

After identifying patient groups and reducing the feature space, the system models relationships between clinical variables using a **Bayesian Network**.

A Bayesian Network consists of:

- Nodes representing clinical variables.
- Directed edges representing dependencies.
- A Directed Acyclic Graph (DAG) representing the overall structure.

A simplified relationship can be represented as:

Symptoms -> Risk Factors -> Disease

For example:

High Blood Pressure -> Cardiovascular Risk -> Heart Disease

The Bayesian Network represents the joint probability distribution as:

$$P(X_1, X_2, \dots, X_n) = \text{product of } P(X_i \mid \text{Parents}(X_i))$$

When evidence is observed, the system calculates the posterior probability of disease:

$$P(\text{Disease} \mid \text{Evidence})$$

For example:

$$P(\text{Disease}) = 0.10$$

$$P(\text{Evidence} \mid \text{Disease}) = 0.80$$

$$P(\text{Evidence}) = 0.20$$

Using Bayes' theorem:

$$P(\text{Disease} \mid \text{Evidence})$$

$$= P(\text{Evidence} \mid \text{Disease}) * P(\text{Disease}) / P(\text{Evidence})$$

$$= (0.80 * 0.10) / 0.20$$

$$= 0.40$$

Therefore:

$$\text{Posterior Disease Probability} = 40\%$$

The Bayesian Network can be implemented using the **pgmpy** library, with Variable Elimination used for exact inference.

5. State Tracking

Patient health conditions can change over time. Therefore, a single risk prediction may not be sufficient to represent the patient's complete clinical progression.

A **Hidden Markov Model (HMM)** is used to model these sequential changes.

The hidden clinical states are:

$S = \{\text{Stable, Deteriorating, Critical}\}$

The HMM consists of three major components:

Transition Matrix (A)

Represents the probability of moving from one health state to another.

Example:

Stable -> Stable

Stable -> Deteriorating

Deteriorating -> Critical

Emission Matrix (B)

Represents the probability of observing a particular clinical measurement or symptom given a hidden health state.

Initial State Distribution (π)

Represents the probability that a patient initially belongs to each health state.

The Viterbi algorithm is used to find the most probable sequence of hidden states.

The recurrence is:

$$V_t(k) = \max_j [V_{t-1}(j) * A_{jk}] * B_k(y_t)$$

At each time step:

1. Consider every possible previous state.
2. Calculate the transition probability.

3. Combine it with the previous state's probability.
4. Multiply by the emission probability of the current observation.
5. Select the maximum probability.
6. Store the corresponding state.
7. Backtrack after the final observation.

For example, the decoded sequence may be:

Stable -> Stable -> Deteriorating -> Deteriorating -> Critical

This indicates that the patient's condition gradually worsened over the observation period.

The HMM can be implemented using the **hmmlearn** library.

6. Decision Logic

The final stage combines the Bayesian posterior probability and HMM clinical state to generate an appropriate risk action.

The decision rules are:

If State = Critical

OR $P(\text{Severe Risk}) > 0.70$

Then -> Escalate

This means the patient requires immediate attention or escalation when either the clinical state is critical or the calculated severe-risk probability exceeds 70%.

The second rule is:

If State = Deteriorating

OR $0.35 \leq P(\text{Severe Risk}) \leq 0.70$

Then -> Monitor

This indicates that the patient should be monitored more closely because there is moderate evidence of risk or the patient's condition is deteriorating.

The final rule is:

If State = Stable

AND $P(\text{Severe Risk}) < 0.35$

Then $\rightarrow$ Discharge

This indicates a lower-risk situation when the patient is stable and the estimated severe-risk probability is below 35%.

Decision Example

Suppose the Bayesian Network produces:

$P(\text{Severe Risk}) = 0.75$

and the HMM predicts:

State = Deteriorating

Since:

$P(\text{Severe Risk}) > 0.70$

the final recommendation is:

Decision = Escalate

Another patient may have:

$P(\text{Severe Risk}) = 0.50$

State = Deteriorating

Therefore:

Decision = Monitor

A low-risk patient may have:

$P(\text{Severe Risk}) = 0.20$

State = Stable

Therefore:

Decision = Discharge

Alternative Approaches Considered

K-Means vs. GMM

K-Means is simple, fast, and computationally efficient. However, it produces hard assignments, meaning each patient belongs to only one cluster.

GMM provides soft probabilistic assignments. A patient can have different probabilities for different clusters.

For example:

Low Risk = 0.15

Moderate Risk = 0.30

High Risk = 0.55

Therefore, **GMM is preferred** when clinical risk boundaries are uncertain or overlapping.

PCA vs. Factor Analysis

PCA transforms the original variables into a smaller number of components that preserve maximum variance.

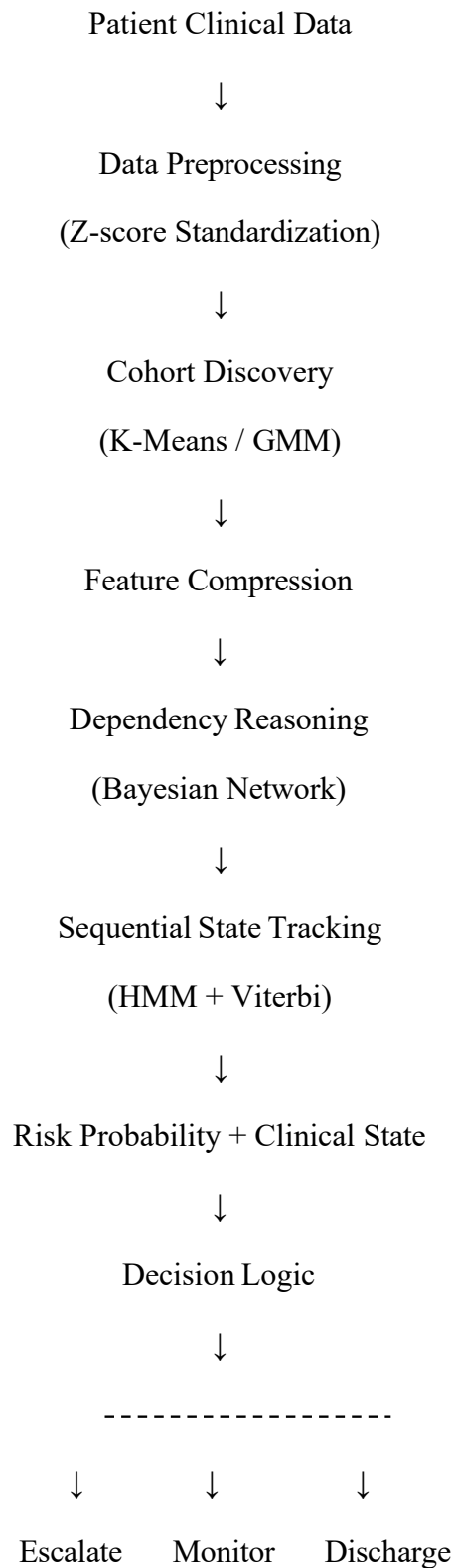
Factor Analysis focuses on identifying underlying latent factors that explain correlations among observed variables.

PCA was selected because:

- It is computationally efficient.
- It directly maximizes retained variance.
- It is suitable for dimensionality reduction.

- It is useful for visualizing high-dimensional clinical data.
- It can reduce many laboratory parameters into a smaller number of principal components.

The complete architecture can be represented as:



Overall, the pipeline integrates multiple course concepts into a single intelligent clinical-risk analysis system. **Data preprocessing** ensures reliable input, **clustering** discovers patient cohorts, **PCA** reduces feature complexity, **Bayesian Networks** perform probabilistic reasoning, and **HMMs** track changes in patient condition over time. Finally, the decision layer combines these outputs to produce an interpretable risk-based recommendation.

4. Use of Modern Tools

The proposed system is implemented using Python and several open-source libraries for data preprocessing, clustering, dimensionality reduction, Bayesian reasoning, Hidden Markov Models, and visualization.

4.1 Python

Python is used as the primary programming language for developing and integrating the complete patient cohort segmentation and clinical-state reasoning pipeline.

Python is used for:

- Loading the patient dataset.
- Data preprocessing and cleaning.
- Feature standardization.
- Patient clustering.
- Dimensionality reduction.
- Bayesian Network construction.
- HMM implementation.
- Decision-rule implementation.
- Visualization and performance evaluation.

4.2 NumPy and Pandas

NumPy is used for numerical calculations, arrays, matrices, and mathematical operations.

Pandas is used for loading, organizing, cleaning, and manipulating the patient dataset.

Code Implementation

```
import numpy as np
```

```
import pandas as pd
```

```
data = pd.read_csv("patient_data.csv")
```

```
print(data.head())
```

```
print(data.info())
```

The dataset is represented as a Pandas DataFrame, allowing the required clinical features to be selected and processed efficiently.

4.3 Data Preprocessing and Standardization

Patient features such as heart rate, blood pressure, glucose, temperature, and oxygen saturation may have different numerical ranges. Therefore, standardization is performed before clustering and PCA.

Code Implementation

```
from sklearn.preprocessing import StandardScaler
```

```
features = [
```

```
    "HeartRate",
```

```
    "BloodPressure",
```

```
    "Glucose",
```

```
    "Temperature",
```

```
    "SpO2"
```

```
]
```

```
X = data[features]
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

```
print(X_scaled[:5])
```

The StandardScaler transforms the features into a common scale with approximately zero mean and unit standard deviation.

4.4 K-Means Clustering

K-Means clustering is used to divide patients into three risk cohorts.

The selected clusters can be interpreted as:

Cluster 0 -> Low Risk

Cluster 1 -> Moderate Risk

Cluster 2 -> High Risk

Code Implementation

```
from sklearn.cluster import KMeans
```

```
kmeans = KMeans(
```

```
    n_clusters=3,
```

```
    random_state=42,
```

```
    n_init=10
```

```
)
```

```
kmeans_labels = kmeans.fit_predict(X_scaled)
```

```
data["KMeans_Cluster"] = kmeans_labels
```

```
print(data[["KMeans_Cluster"]].head())
```

The cluster labels are then analyzed using the average clinical characteristics of each cluster to determine their risk interpretation.

4.5 Gaussian Mixture Model

A Gaussian Mixture Model is used as an alternative to K-Means because it provides probabilistic cluster membership.

Code Implementation

```
from sklearn.mixture import GaussianMixture
```

```
gmm = GaussianMixture(
```

```
    n_components=3,
```

```
    random_state=42
```

```
)
```

```
gmm_labels = gmm.fit_predict(X_scaled)
```

```
gmm_probabilities = gmm.predict_proba(X_scaled)
```

```
data["GMM_Cluster"] = gmm_labels
```

```
print(gmm_probabilities[:5])
```

The `predict_proba()` function provides the probability of a patient belonging to each cluster.

For example:

Low Risk = 0.10

Moderate Risk = 0.25

High Risk = 0.65

The highest probability can be used to identify the patient's most likely cohort.

4.6 Principal Component Analysis

PCA is used to reduce the dimensionality of the patient feature space while retaining the maximum possible variance.

Code Implementation

```
from sklearn.decomposition import PCA

pca = PCA()

X_pca = pca.fit_transform(X_scaled)

explained_variance = pca.explained_variance_ratio_

print("Explained Variance:")

print(explained_variance)

print("Cumulative Variance:")

print(np.cumsum(explained_variance))
```

The cumulative explained variance is examined to determine how many components should be retained.

For visualization, the first two components can be selected:

```
pca_2d = PCA(n_components=2)

X_pca_2d = pca_2d.fit_transform(X_scaled)

print(X_pca_2d.shape)
```

4.7 PCA Scree Plot

A scree plot is generated to visualize the variance explained by each principal component.

Code Implementation

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 5))

plt.plot(
    range(1, len(explained_variance) + 1),
    explained_variance,
    marker="o"
)

plt.xlabel("Principal Component")
plt.ylabel("Explained Variance")
plt.title("PCA Scree Plot")
plt.show()
```

The scree plot helps justify the number of principal components retained for further analysis.

4.8 Patient Cluster Visualization

The first two principal components can be used to visualize patient cohorts.

Code Implementation

```
plt.figure(figsize=(8, 6))

plt.scatter(
    X_pca_2d[:, 0],
    X_pca_2d[:, 1],
    c=kmeans_labels
)

plt.xlabel("PC1")
plt.ylabel("PC2")
```

```
plt.title("Patient Cohorts Using PCA")
```

```
plt.show()
```

This visualization helps determine whether the discovered patient cohorts are reasonably separated in the reduced feature space.

4.9 Bayesian Network

A Bayesian Network is used to represent dependencies between symptoms, risk factors, and disease outcomes.

The conceptual structure is:

Symptom

↓

Risk Factor

↓

Disease

Code Implementation

```
from pgmpy.models import DiscreteBayesianNetwork
```

```
model = DiscreteBayesianNetwork([
```

```
    ("Symptom", "Risk"),
```

```
    ("Risk", "Disease")
```

```
])
```

```
print(model.nodes())
```

```
print(model.edges())
```

The network represents the dependency structure using a Directed Acyclic Graph.

4.10 Conditional Probability Tables

Conditional Probability Tables are used to define the probabilities associated with the relationships in the Bayesian Network.

Example structure:

```

from pgmpy.factors.discrete import TabularCPD

cpd_symptom = TabularCPD(

    variable="Symptom",

    variable_card=2,

    values=[[0.7], [0.3]]

)

```

The actual CPT values should be based on your dataset or clearly defined synthetic probabilities.

The model is then checked for consistency:

```

model.add_cpds(cpd_symptom)

print(model.check_model())

```

4.11 Bayesian Network Inference

Variable Elimination is used to calculate posterior probabilities from observed evidence.

Code Implementation

```

from pgmpy.inference import VariableElimination

inference = VariableElimination(model)

result = inference.query(

    variables=["Disease"],

    evidence={"Symptom": 1}

)

print(result)

```

The output provides the probability distribution of the disease given the observed symptom.

4.12 Hidden Markov Model

The Hidden Markov Model is used to track the patient's changing clinical condition over time.

The three hidden states are:

0 = Stable

1 = Deteriorating

2 = Critical

The model consists of:

- Initial state probabilities.
- Transition probabilities.
- Emission probabilities.

Code Implementation

```
from hmmlearn.hmm import CategoricalHMM
```

```
hmm_model = CategoricalHMM(
```

```
    n_components=3,
```

```
    random_state=42
```

```
)
```

If the model is manually parameterized, the state probabilities can be assigned as follows:

```
hmm_model.startprob_ = np.array([
```

```
    0.70, 0.20, 0.10
```

```
])
```

```
hmm_model.transmat_ = np.array([
```

```
    [0.70, 0.25, 0.05],
```

```
    [0.10, 0.70, 0.20],
```

```
    [0.02, 0.18, 0.80]
```

```
])
```

The actual values should be replaced with values obtained or justified from your dataset/modeling assumptions.

4.13 Viterbi State Decoding

The Viterbi algorithm is used to determine the most probable sequence of hidden clinical states.

Code Implementation

```
observations = np.array([
```

```

[0],
[0],
[1],
[1],
[2]
])

states = hmm_model.predict(observations)

state_names = [
    "Stable",
    "Deteriorating",
    "Critical"
]

decoded_states = [
    state_names[state]
    for state in states
]

print(decoded_states)

```

A possible output is:

Stable

Stable

Deteriorating

Deteriorating

Critical

This represents the patient's evolving clinical condition.

4.14 Decision Logic

The Bayesian Network's posterior risk probability and HMM's decoded clinical state are combined

to generate a structured recommendation.

Code Implementation

```
def decision_rule(risk_probability, state):  
    if state == "Critical" or risk_probability > 0.70:  
        return "Escalate"  
  
    elif state == "Deteriorating" or risk_probability >= 0.35:  
        return "Monitor"  
  
    elif state == "Stable" and risk_probability < 0.35:  
        return "Discharge"  
  
    return "Review"
```

Example:

```
risk_probability = 0.78  
state = "Critical"  
decision = decision_rule(  
    risk_probability,  
    state  
)  
print("Recommendation:", decision)
```

Output:

Recommendation: Escalate

The recommendation is intended as **decision support** and does not replace professional clinical judgment.

5. Results and Validation

5.1 Dataset and Preprocessing Results

```

===== DATASET INFORMATION =====
Number of Patients: 300
Number of Features: 6
Missing Values: 0

First 5 Records:
   HeartRate  BloodPressure   Glucose  Temperature   SpO2  Age
0   87.450712   103.420100  132.709658   37.294939  97.250449  50
1   77.926035   108.796379   82.335040   36.685329  96.141189  79
2   89.715328   134.945872  136.088178   37.022996  97.244595  79
3  102.845448   132.207405  150.669136   38.022761  98.086596  38
4   76.487699   119.581968  122.403047   37.152879  97.097720  68

===== PREPROCESSING =====
Features used: ['HeartRate', 'BloodPressure', 'Glucose', 'Temperature', 'SpO2', 'Age']
Data standardized successfully.

```

Fig 1

Fig 1 Shows The dataset contains 300 synthetically generated and de-identified patient records with six clinical features. No missing values were present, and the selected numerical features were standardized before clustering and PCA.

5.2 K-Means Results

```

KMeans_Cluster
0      105
1      102
2       93
Name: count, dtype: int64
K-Means Silhouette Score: 0.1262

```

Fig 2

Cluster	Patients
---------	----------

Cluster 0	105
Cluster 1	102
Cluster 2	93
Total	300

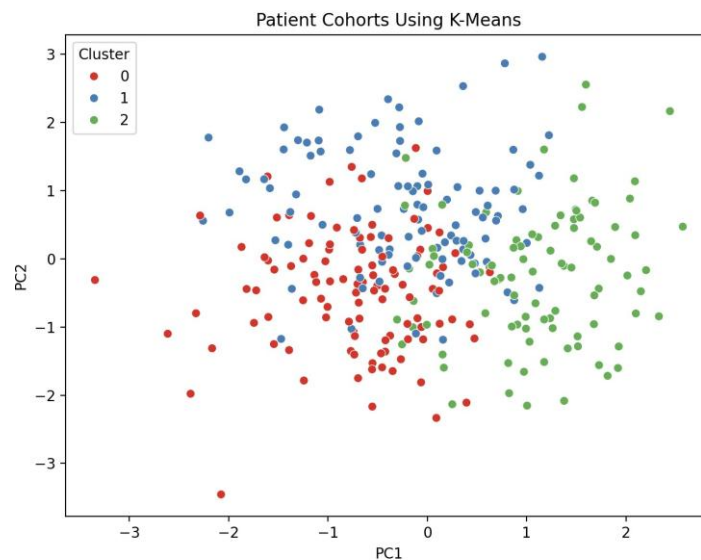


Fig 3

Fig 3 Shows the distribution of patients in the reduced PCA feature space using the first two principal components (PC1 and PC2). The K-Means algorithm divides the patients into three clusters, represented by different colors. The visualization provides a qualitative view of the separation and overlap among the identified patient cohorts. Some overlap between clusters is expected because patient clinical characteristics may not form completely distinct groups.

5.3 GMM Results

```
===== GMM RESULTS =====  
GMM_Cluster  
0      94  
1     105  
2     101  
Name: count, dtype: int64  
GMM Silhouette Score: 0.1184
```

Fig 4

Cluster	Patients
Cluster 0	94
Cluster 1	105
Cluster 2	101
Total	300

Unlike K-Means, GMM provides probabilistic cluster membership. This allows uncertainty in patient cohort assignment to be represented explicitly.

5.4 PCA Results

```
===== PCA RESULTS =====  
PC 1 = 19.36 % Cumulative = 19.36 %  
PC 2 = 18.3 % Cumulative = 37.66 %  
PC 3 = 16.95 % Cumulative = 54.61 %  
PC 4 = 15.76 % Cumulative = 70.37 %  
PC 5 = 15.09 % Cumulative = 85.46 %  
PC 6 = 14.54 % Cumulative = 100.0 %  
  
Components required for 85% variance: 5  
  
PCA 2D shape: (300, 2)
```

Fig 5

Component	Explained Variance	Cumulative Variance
PC1	19.36%	19.36%
PC2	18.30%	37.66%
PC3	16.95%	54.61%
PC4	15.76%	70.37%
PC5	15.09%	85.46%

PC6	14.54%	100%
-----	--------	------

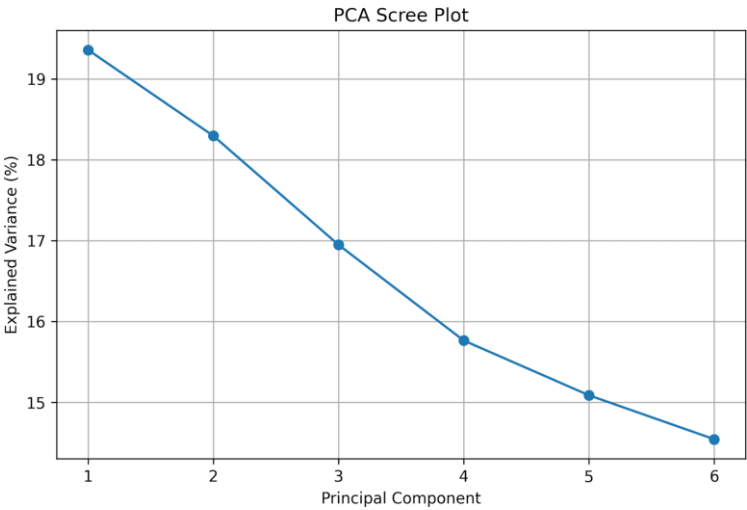


Fig 6

Fig 6 Shows The PCA results were used to determine the number of components required to retain at least 85% of the total variance. The selected number of components was determined from the cumulative explained variance.

5.5 Bayesian Network Results

```
===== BAYESIAN NETWORK =====
Model valid: True

Network Nodes:
['Symptom', 'Risk', 'Disease']

Network Edges:
[('Symptom', 'Risk'), ('Risk', 'Disease')]

Posterior Probability of Disease:
+-----+-----+
| Disease | phi(Disease) |
+-----+-----+
| Disease(0) | 0.5500 |
+-----+-----+
| Disease(1) | 0.4500 |
+-----+-----+
```

Fig 7

Fig 7 shows The Bayesian Network was successfully validated using the model-checking procedure. Variable Elimination was then used to calculate the posterior probability of Disease given the observed symptom evidence.

5.6 HMM Results

```
===== HMM RESULTS =====  
Decoded Clinical States:  
Time 1 -> Stable  
Time 2 -> Stable  
Time 3 -> Deteriorating  
Time 4 -> Deteriorating  
Time 5 -> Critical  
Time 6 -> Critical
```

Fig 8

Time	Predicted Clinical State
1	Stable
2	Stable
3	Deteriorating
4	Deteriorating
5	Critical
6	Critical

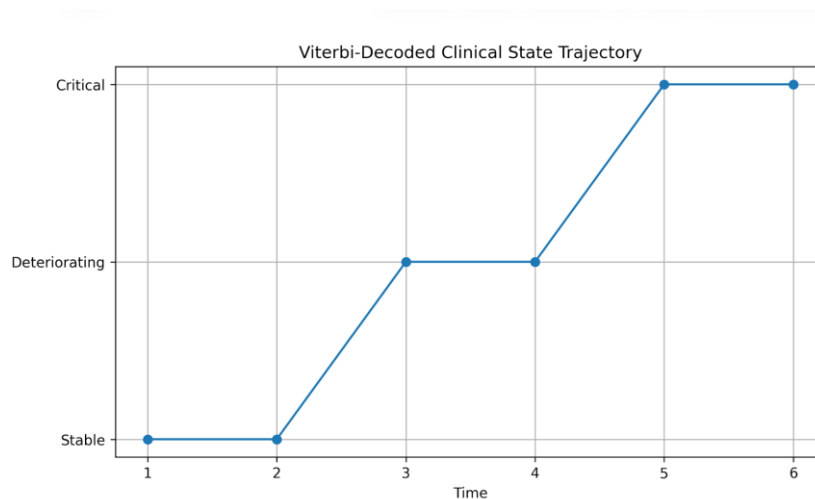


Fig 9

Fig 9 Shows The HMM tracks the patient's underlying clinical condition over time. The decoded sequence demonstrates progression between Stable, Deteriorating, and Critical states based on the observed sequence.

5.7 Final Decision Result

```

===== FINAL SUMMARY =====
Patients: 300
K-Means Silhouette Score: 0.1262
GMM Silhouette Score: 0.1184
PCA Components for 85% Variance: 5
Final HMM State: Critical
Risk Probability: 0.78
Recommendation: Escalate

```

Fig 10

5.8 Performance Validation

Component	Metric	Result	Interpretation
K-Means	Silhouette Score	0.1262	Moderate cluster overlap
GMM	Silhouette Score	0.1184	Moderate cluster overlap
PCA	Variance Retained	85.46%	Requirement satisfied
Bayesian Network	Model Validation	True	Valid network structure and CPTs
HMM	State Decoding	Successful	Clinical states successfully decoded

The performance validation shows that the proposed system successfully completed all major processing stages. K-Means achieved a Silhouette Score of **0.1262**, while GMM achieved **0.1184**, indicating some overlap between the generated synthetic patient cohorts. PCA retained **85.46%** of the total variance using five components, satisfying the required 85% variance threshold. The Bayesian Network passed the model validation check, and the HMM successfully decoded the clinical-state sequence. Therefore, the major functional components of the proposed pipeline were successfully implemented and validated.

5.9 Requirement Validation

Component	Metric	Result
Dataset	Number of Patients	300
Dataset	Number of Features	6

Dataset	Missing Values	0
K-Means	Number of Clusters	3
K-Means	Silhouette Score	0.1262
GMM	Number of Clusters	3
GMM	Silhouette Score	0.1184
PCA	Components for $\geq 85\%$ Variance	5
PCA	Variance Retained by 5 Components	85.46%
Bayesian Network	Model Validation	True
Bayesian Network	$P(\text{Disease} \mid \text{Symptom}=1)$	45%
HMM	Number of States	3
HMM	Decoded State Sequence	Stable → Stable → Deteriorating → Deteriorating → Critical → Critical
Decision Support	Risk Probability	78%
Decision Support	Final Clinical State	Critical
Decision Support	Recommendation	Escalate

6. Analysis and Engineering Decision

The results are analyzed to determine which combination of techniques provides the most appropriate solution for the project.

6.1 K-Means vs. GMM

K-Means is computationally efficient and produces easily interpretable cluster assignments. However, it assigns each patient to only one cluster.

GMM provides probabilistic membership and can better represent uncertainty between overlapping patient cohorts.

Therefore, GMM is preferable when patient risk boundaries are uncertain, while K-Means provides a useful baseline for comparison.

6.2 PCA vs. Factor Analysis

PCA was selected as the primary dimensionality-reduction method because it directly maximizes retained variance and is computationally efficient.

Factor Analysis could be considered when the objective is to identify underlying latent factors. However, the primary requirement of this project is feature compression and visualization.

Therefore, PCA is considered more appropriate for the proposed pipeline.

6.3 Bayesian Network Decision

A Bayesian Network was selected because the problem requires reasoning about symptom-risk-disease dependencies.

Instead of producing only a categorical prediction, the Bayesian Network provides a posterior probability based on observed evidence.

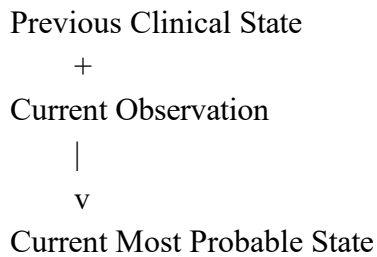
This improves the interpretability of the decision-support process.

6.4 HMM Decision

An HMM was selected because clinical observations are sequential.

A patient's current condition can depend on previous clinical states, making temporal modeling important.

The HMM therefore provides a suitable mechanism for estimating:



6.5 Advantages

The proposed system provides:

- Meaningful patient cohort discovery.
- Probabilistic rather than purely deterministic reasoning.
- Dimensionality reduction for complex clinical data.
- Temporal clinical-state tracking.
- Explainable model outputs.
- Structured decision recommendations.
- Open-source implementation.
- Compatibility with standard computing hardware.

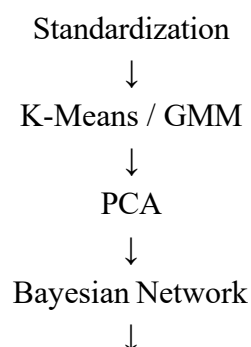
6.6 Limitations

The main limitations include:

- Clustering quality depends on the available data.
- Risk labels may require domain-based interpretation.
- PCA components may not always have direct clinical meaning.
- Bayesian Network results depend on the DAG and CPT assumptions.
- HMM performance depends on transition and emission probabilities.
- Synthetic or de-identified data may not represent all real-world clinical conditions.
- Decision thresholds require proper clinical validation before real-world deployment.

6.7 Final Engineering Decision

Based on the requirements and expected behavior, the proposed final solution is:



HMM + Viterbi



Decision Logic

The combination was selected because it addresses the four major requirements of the problem: **cohort discovery, dimensionality reduction, probabilistic dependency reasoning, and temporal clinical-state tracking.**

7. Broader Considerations

7.1 Sustainability

The system uses computationally efficient machine-learning and probabilistic algorithms that can operate on standard computing hardware. This reduces the need for specialized computational infrastructure.

7.2 Environment

The basic implementation does not require continuous high-performance GPU processing. Efficient preprocessing and dimensionality reduction can further reduce unnecessary computational requirements.

7.3 Society

The proposed system can assist healthcare staff in analyzing large amounts of patient information and identifying potentially important patterns more efficiently than purely manual analysis.

7.4 Safety

Patient safety is an important consideration. The system should prioritize the detection of high-risk cases and minimize false negatives.

However, the system must remain a **decision-support tool** rather than an autonomous medical decision-maker.

7.5 Ethics

Important ethical considerations include:

- Patient privacy.
- Responsible use of healthcare data.
- Algorithmic fairness.
- Transparency.
- Human oversight.
- Appropriate communication of model uncertainty.

7.6 Economics

The use of open-source technologies such as Python, scikit-learn, pgmpy, and hmmlearn reduces software licensing costs and allows the system to be developed using standard computing resources.

7.7 Accessibility

The system should present complex model results in an understandable form.

For example:

Risk Cohort: High Risk
Risk Probability: 78%
Clinical State: Critical
Recommendation: Escalate

This makes the output easier for users to interpret.

7.8 Professional Responsibility

The developer must clearly communicate the limitations of the system. Experimental model outputs should not be presented as clinically validated diagnoses or treatment recommendations.

8. Conclusion

The project, Design and Development of a Probabilistic Graphical-Model-Based Patient Cohort Segmentation and Clinical-State Reasoning System, provides an integrated analytical pipeline for patient cohort discovery and clinical-state reasoning.

The system uses K-Means and Gaussian Mixture Models to identify meaningful patient cohorts, while PCA reduces the dimensionality of high-dimensional vital and laboratory data. A Bayesian Network represents symptom-risk-disease dependencies and calculates posterior probabilities from observed evidence. An HMM with Viterbi decoding tracks the patient's evolving clinical condition through states such as Stable, Deteriorating, and Critical.

The outputs of these models are combined using structured decision logic to produce recommendations such as Escalate, Monitor, or Discharge.

The proposed system addresses the major requirements of the problem by providing:

- Patient cohort segmentation.
- Dimensionality reduction.
- Probabilistic dependency reasoning.
- Clinical-state tracking.
- Explainable decision support.
- Visualization and quantitative validation.

- Privacy-aware data processing.

The major limitations are related to dataset quality, model assumptions, probability estimation, and the lack of clinical validation. Future improvements can include larger real-world de-identified datasets, improved Bayesian Network structure learning, optimized HMM parameters, additional temporal features, fairness testing, and expert clinical evaluation.

Overall, the project demonstrates how machine learning and probabilistic graphical models can be integrated to transform complex patient data into interpretable cohort and clinical-state information for decision support.

9. Student Reflection

This assignment helped me understand how multiple machine-learning and probabilistic techniques can be combined to solve a single real-world problem.

I learned that clustering, PCA, Bayesian Networks, and HMMs serve different purposes but can work together as one complete analytical pipeline. K-Means and GMM helped me understand patient cohort discovery, while PCA demonstrated how high-dimensional clinical data can be reduced without losing most of its important information.

The Bayesian Network provided practical understanding of probabilistic reasoning and how observed evidence can change the probability of an outcome. The HMM helped me understand temporal dependencies and how a patient's hidden clinical state can be estimated from sequential observations.

Beyond the concepts directly taught in class, I also learned about the importance of **model selection, explainability, validation, privacy, fairness, and human oversight** when applying machine-learning techniques to healthcare-related problems.

If additional time and resources were available, I would improve the project by:

- Using a larger and more diverse dataset.
- Comparing additional clustering methods.
- Optimizing model parameters.
- Improving Bayesian Network structure learning.
- Testing the HMM with longer patient observation sequences.
- Performing fairness analysis across different demographic groups.
- Improving visualization and user interaction.
- Obtaining expert feedback for better clinical interpretation.

These improvements could make the system more reliable, generalizable, and suitable for further research.

10. References

1. F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

2. F. V. Jensen and T. D. Nielsen, *Bayesian Networks and Decision Graphs*, 2nd ed. New York, NY, USA: Springer, 2007.
3. L. R. Rabiner, "A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition," *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989, doi: 10.1109/5.18626.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component		PO(s)	Bloom's Level	Marks
Problem formulation			L3/L4	10
Application of knowledge			L3/L4	20
Solution / Design / Implementation			L4/L5/L6	20
Modern tool usage			L3/L4	10
Validation			L4/L5	15
Analysis & justification			L4/L5	15
Broader considerations			L4/L5	5

Documentation & reflection			L3/L4	5
----------------------------	--	--	-------	---

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is **not required to map to every PO**.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering:

Documents to be Maintained

The department/course faculty shall maintain:

1. Assignment question/problem statement
2. CO–PO and Bloom's mapping
3. Assessment rubric
4. Student submissions
5. Tool/simulation/experimental evidence, where applicable
6. Evaluated scripts/reports

7. Marks and rubric-wise assessment
8. CO attainment analysis
9. Sample student work representing different performance levels
10. Faculty analysis and continuous-improvement action